GRID-PULSE NTL // Kafr El-Sheikh · autonomous NTL detection

The grid loses power it can't see. We find it.

Every distribution transformer leaks energy — some to heat, some to theft. Grid-Pulse learns how much loss is **normal right now** — load, temperature, hour — and flags only the unexplained rest as theft, with the reason it decided.

≈ 30B EGP

lost to non-technical loss in Egypt, every year

< 1 month

payback per instrumented transformer kiosk

◆ THEFT SUSPECTED TX-KFS-0456 · single

94% theft probability

~3.08 kW unaccounted beyond natural loss — dispatch field inspection.

- ✓ **Energy imbalance** +3076 W
- ✓ **Expected loss exceeded** +951% vs baseline
- ✓ **Power factor dropped** 0.66 (≥ 0.92)
- ✓ **THD increased** 22% (> 10%)
- ✓ **Load outside normal pattern** anomaly 0.83

3.40 kW	0.32 kW	3.08 kW
MEASURED LOSS	AI BASELINE	UNACCOUNTED

01 THE PROBLEM

Why the grid is blind

Cable and transformer losses are real, and they rise with heat and load. A fixed threshold cries wolf all summer — so operators mute it, and real theft hides in the noise.

A brick-kiln, a workshop, a battery of hair-dryers hooked past the meter — the transformer feels it as "more loss." Without context, every hot afternoon looks like theft, and every theft looks like a hot afternoon.

~30%

of generated power lost to NTL in the worst-hit zones

7

independent inputs the model weighs — not temperature alone

3 beats

is all it takes to show a judge the difference, live

Separating heat from theft

Same transformer, three readings. The AI baseline rises with the heat — so the middle case raises **no** alarm. Only the third leaves an unexplained residual. This is the whole demo.

● NORMAL

Baseline load

Registered demand, clean waveform. Energy balance holds.

load	6.85 kW
power factor	0.98
residual	−0.17 kW
probability	0%

● TECHNICAL LOSS

Hot day, 45 °C

Loss climbs — but the model expects it. R rises with heat. No false alarm.

load	7.00 kW
AI baseline	↑ tracks temp
residual	−0.15 kW
probability	0%

● THEFT SUSPECTED

Unregistered hook

+3.4 kW draw, PF collapses, THD spikes. Baseline stays flat — residual explodes.

load	10.2 kW
power factor	0.66
residual	**+3.08 kW**
probability	**94%**

Multi-factor, and it shows its work

A temperature-only model reads as a toy. Ours weighs seven signals; a Random-Forest learns what natural loss should be, and the verdict names every reason it fired.

LEARNED FEATURE IMPORTANCE · TECHNICAL-LOSS MODEL

DETECTION PIPELINE

① **Energy balance**
feeder power − registered meters = measured loss

② **AI technical-loss baseline**
predicts normal loss now → residual = measured − expected

③ **Harmonics & power factor**
illegal loads distort the waveform

④ **Temporal anomaly**
Isolation Forest flags draws that don't fit the curve

⑤ **Fusion + explainable verdict**
probability, reasons checklist, estimated stolen kW

04 THE HARDWARE

From breadboard to busbar

A real edge node, not a slide. Current transformer → signal conditioning → ESP32 → a strict, signed telemetry contract the backend enforces on every packet.

EDGE NODE — TODAY

ESP32 + split-core CT

- SCT-013-000 clamp, 100 A : 50 mA, non-invasive
- 33 Ω burden + 1.65 V bias into the ADC — measured
- EmonLib RMS, calibration ≈ 60.6, NTP-timestamped
- Posts contract-valid JSON with a bearer token

PRODUCTION — ROADMAP

Kiosk-grade sensing

- Rogowski coils on the LV busbar, IP65
- LoRaWAN backhaul — no field Wi-Fi
- TinyML on-device: send only on theft (−99% bw)
- Theft-location triangulation across feeders

BACKEND

FastAPI · Redis · Celery

- Strict JSON-Schema contract: bad body → 422
- Bad token → 401, every packet authenticated
- Inline or worker detection — same verdict
- Live map + verdicts over WebSocket

THE INTERFACE

An operator's console

- Every transformer on a map: green / amber / red
- Big probability + reasons checklist
- One tap: dispatch inspection, or dismiss
- 2 s polling fallback if the socket drops

A business the ministry funds itself

We don't sell software they have to justify — we take a share of the energy we recover. The first month of caught theft pays for the kiosk.

≈ 30B EGP

annual NTL loss in Egypt — the pool we recover against

< 1 mo

ROI per instrumented kiosk at typical recovery rates

SaaS + HaaS

hardware-as-a-service + platform, or % of recovered energy

Customer: distribution companies and the Ministry of Electricity. **Wedge:** one region, a few hundred kiosks, measured recovery. **Moat:** the labelled theft data we accumulate makes every next region's model sharper — a data advantage a fixed-threshold competitor can never catch.

THE PITCH, IN ONE LINE

We turn a blind meter reading into a decision an inspector can act on.

Working edge node, real trained model, explainable verdict, and a demo that lands in three clicks. Built in Kafr El-Sheikh — to be deployed across the grid.

NORMAL · TECHNICAL_LOSS · NTL_THEFT_SUSPECTED